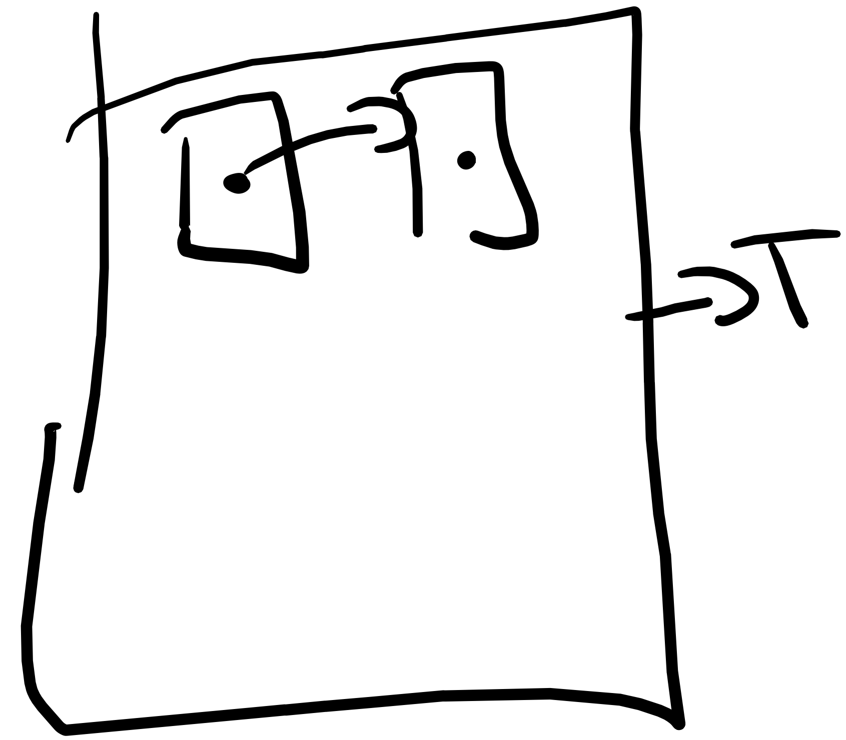
Spatial Processing

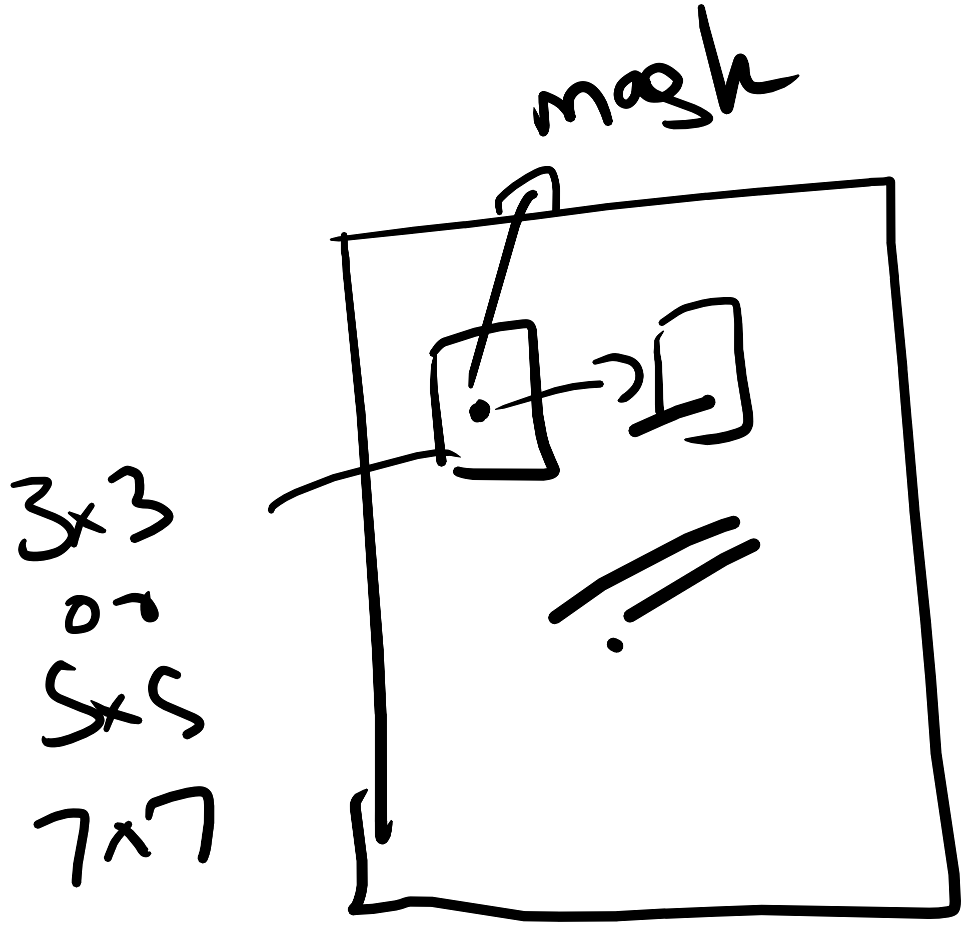
Blurring $\Rightarrow$

masked image

$$x[i,j] = \text{mean}(Z)$$

Spatial filtering





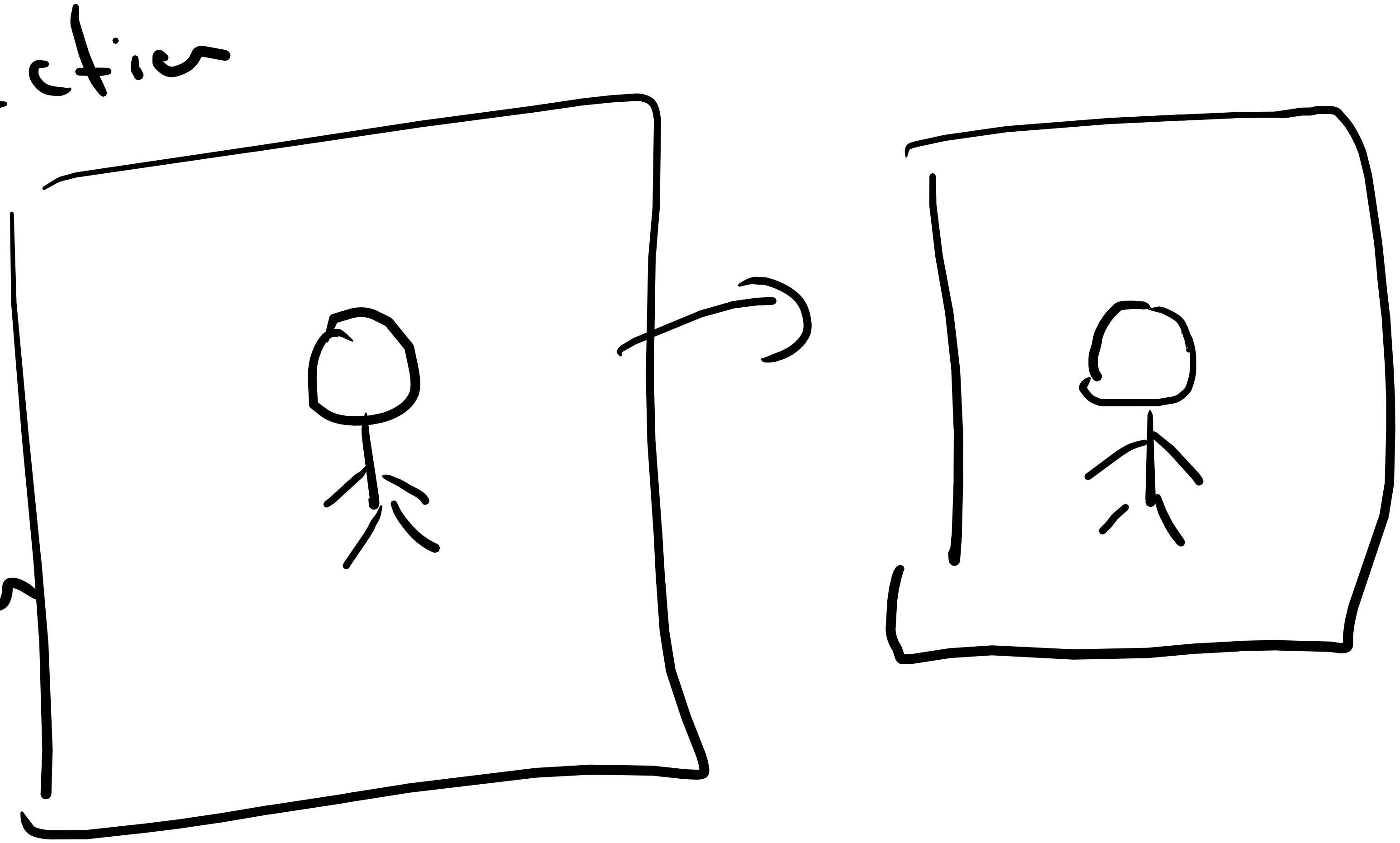
⇒ Spatial filtering

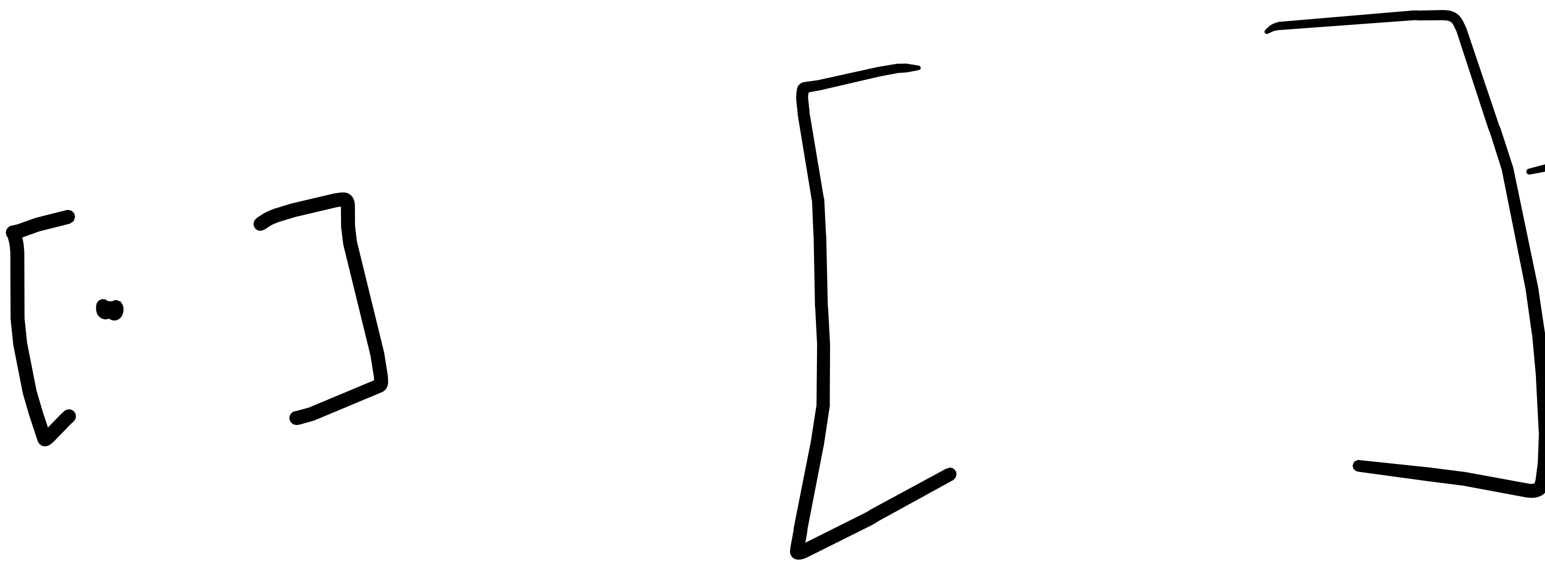
↓

Blurring

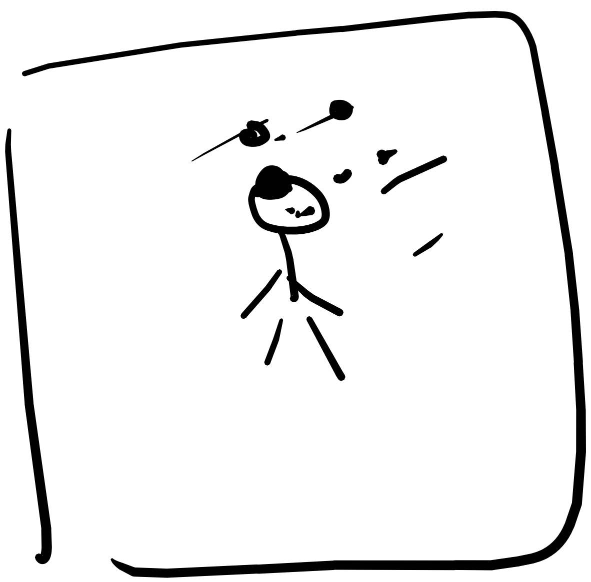
Edge detection

Edge Detection used for obstacle detection





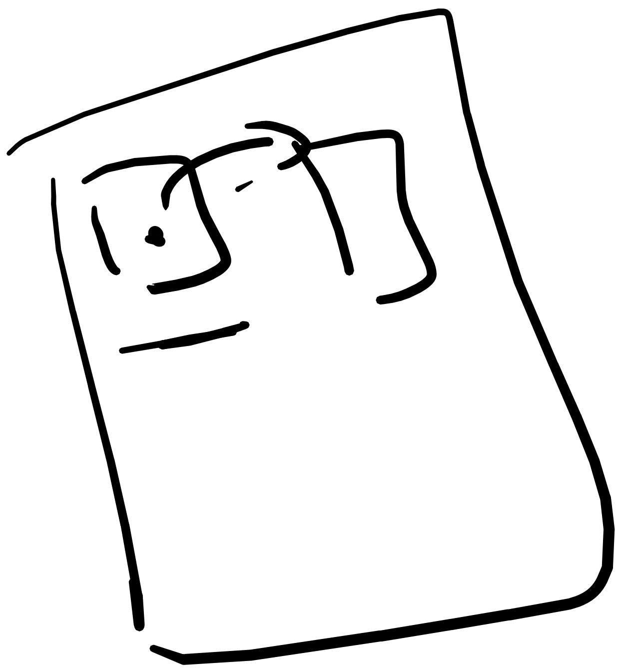
3x3 mask



← In open CV ⇒ well defined functions are available

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} = 0$$

Laplacian filter



filter1 $\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$

filter2

$$\begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \\ w_7 & w_8 & w_9 \end{bmatrix} \Rightarrow \text{LeCun} \Rightarrow \underline{\text{LeNet}}$$

$\Downarrow$

<u>CNN</u>

Color Processing

img = iio.imread(' ')

(m, n)

(m, n, 3)

$\boxed{R, G, B}$ or (Y, M, Cy)

Printers

HSV

Storing image files.

Color Coding

Hue, Saturation and Value

openCV Hue, Saturation and Value. (HSV)

Brightness
or
Intensity

Hue ⇒ Colors
↓
[0, 179]

Red ∈ [0, 10]
Yellow ∈ [20, 30]
Green ∈ [35, 85]

Blue ∈ [100, 130]

(Saturation => Amount of white light mixed in pure color

dark green

green (black)

light green

S=255

S=0

Value ⇒ Intensity

light

dark

scale of intensity

[0, 255]

black

white

grey

Hue, Saturation => exact Color

Value => Intensity

for all Color detection task => first Convert RGB image to HSV.

RGB

↓

fine for

deeplearning

HSV

↓

Color processing

or

human Consumption

Image formation

microscopic images

Vehicles — Cameras are different

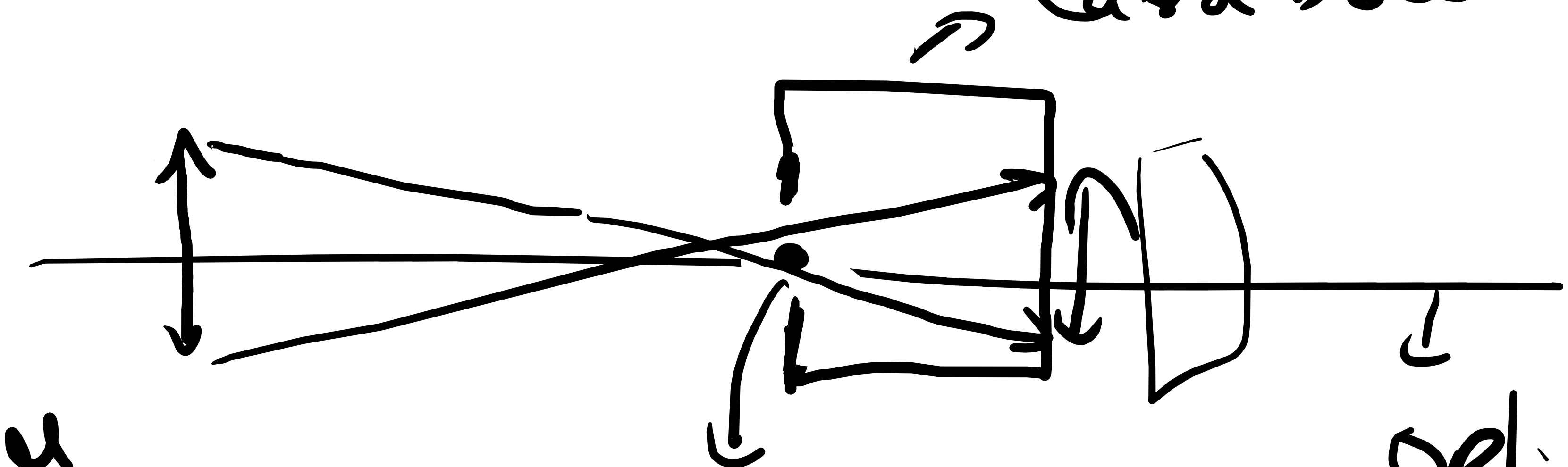
Robots —

Pin-hole Camera

images are projection

Aperture has to be small for image to form.

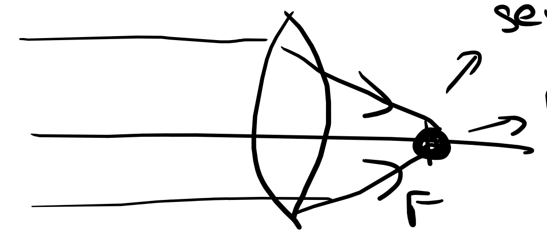
Cardboard



optical axis

=> Difficult to get bright images

Lens : Case 1:

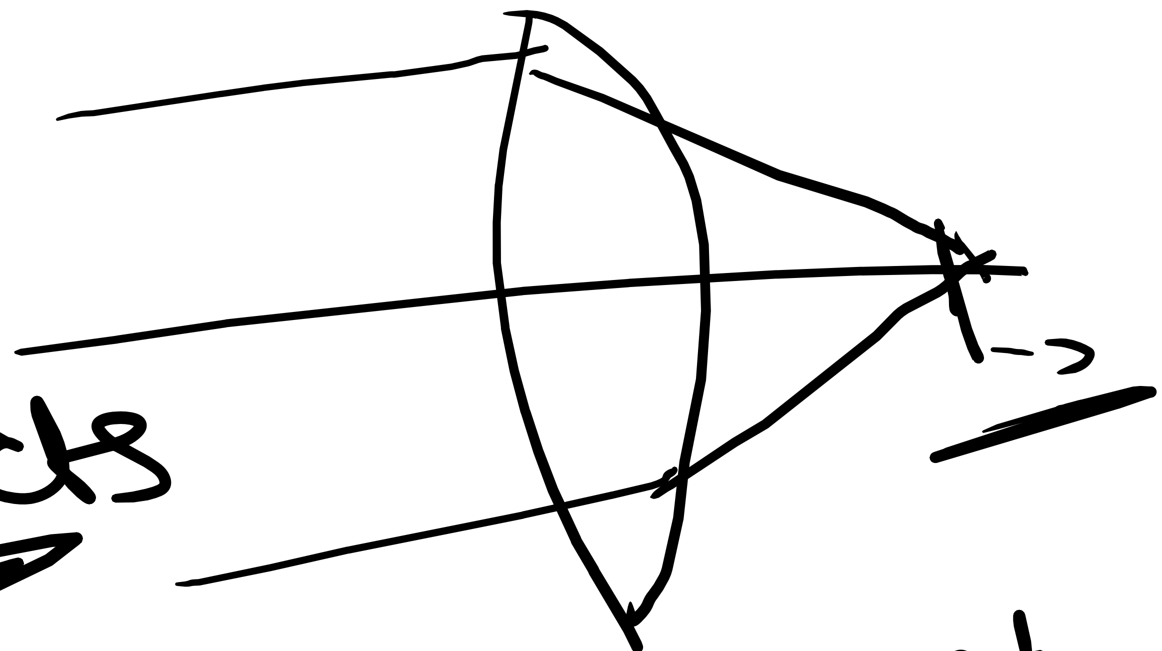


sensor or photographic film at focus.

Point image at the focus

Case 1:

for distant objects



Arrangement of lenses and focal length of lenses

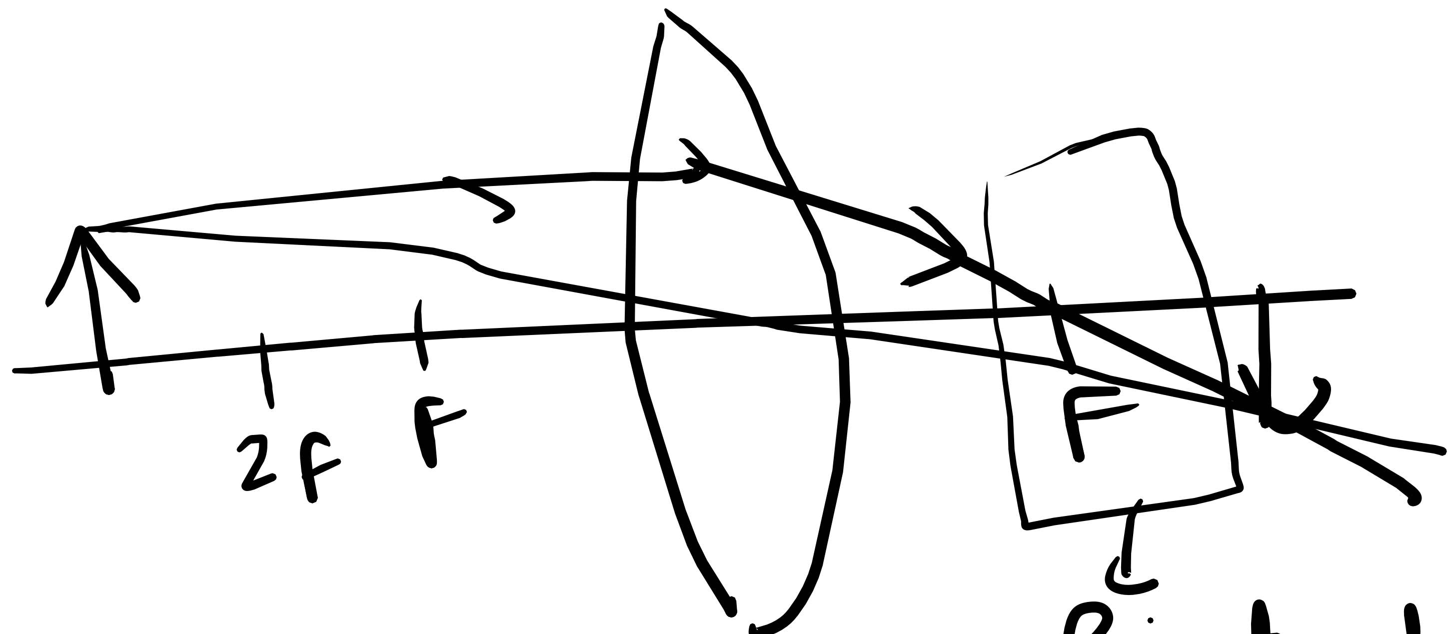
CV Pipeline

Averaging is like removing noise

for lens/optics Correction

Series of DIP (Blurring, noise removal, edge detection

Case 2: Near object (> 2f)



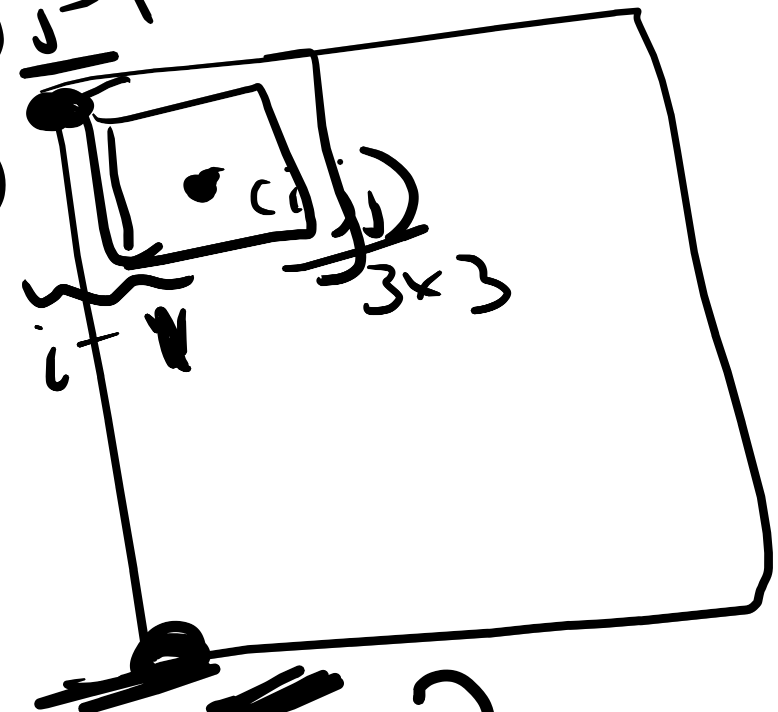
Point where we lose information or image becomes less sharp.

nrow

j=1

i=1

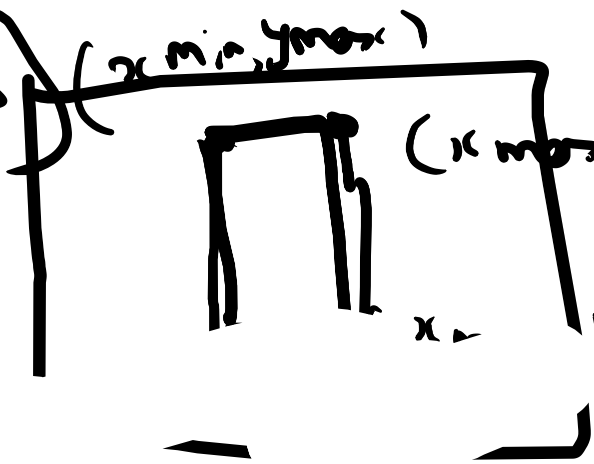
(0,0)



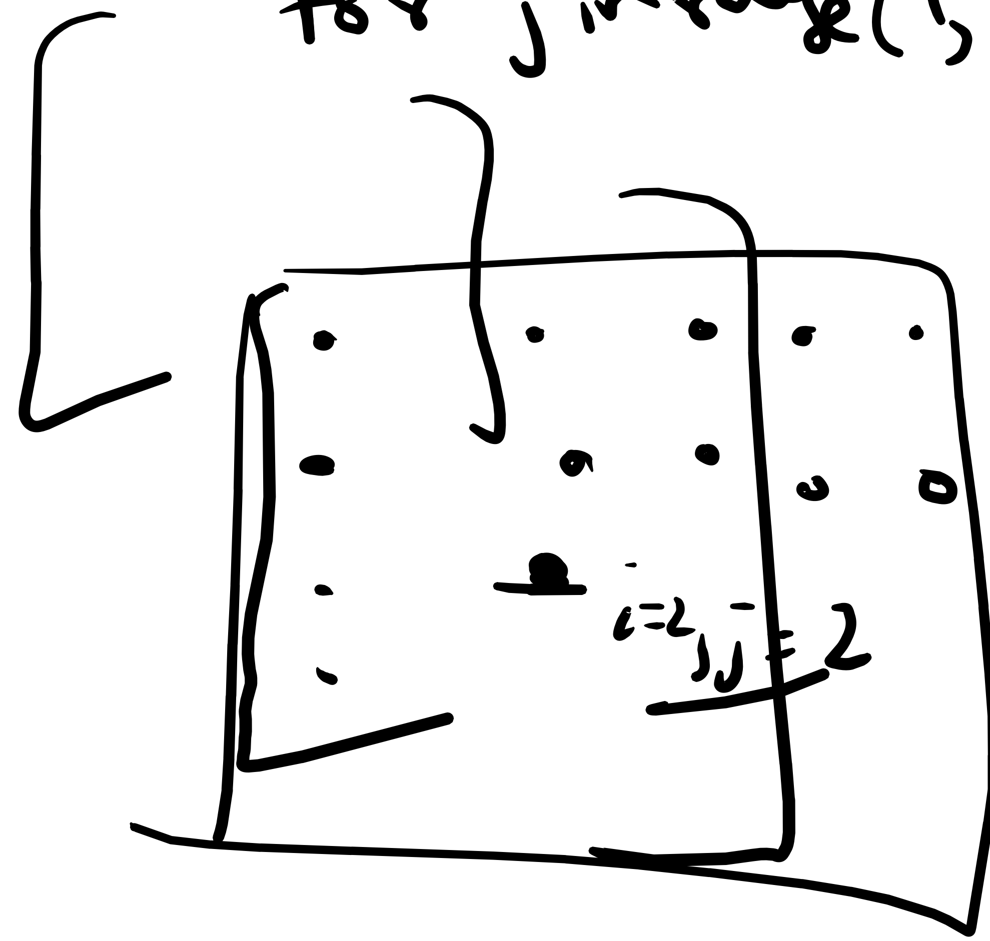
i=n-2

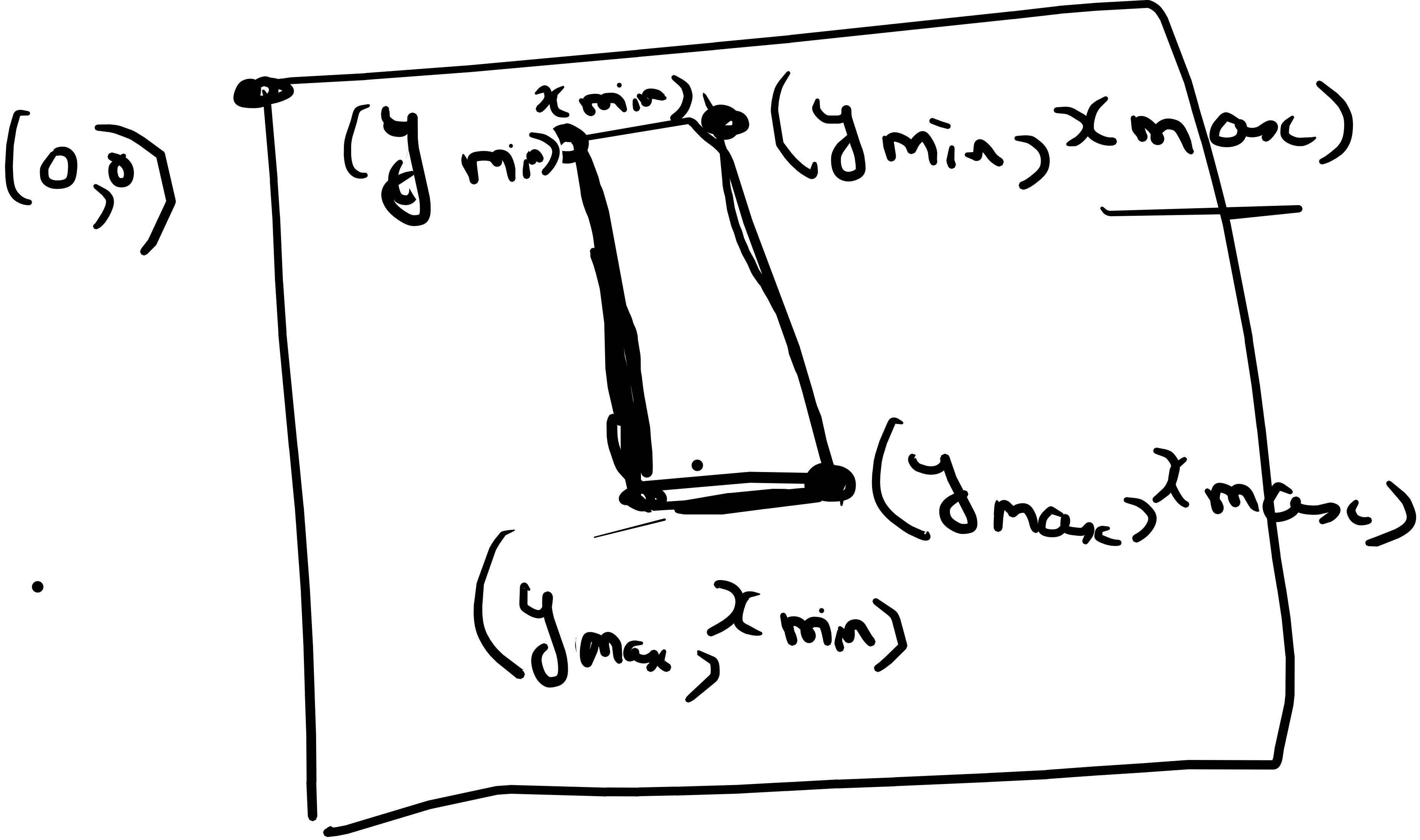
BGR

[0,0,255]



for i in range(1, nrow-1)
for j in range(1, ncol-1)





(0,0)
(ymin, xmin)
(ymin, xmax)
(ymax, xmax)
(ymax, xmin)